Manual's user for the Discrete channel AWGN simulator

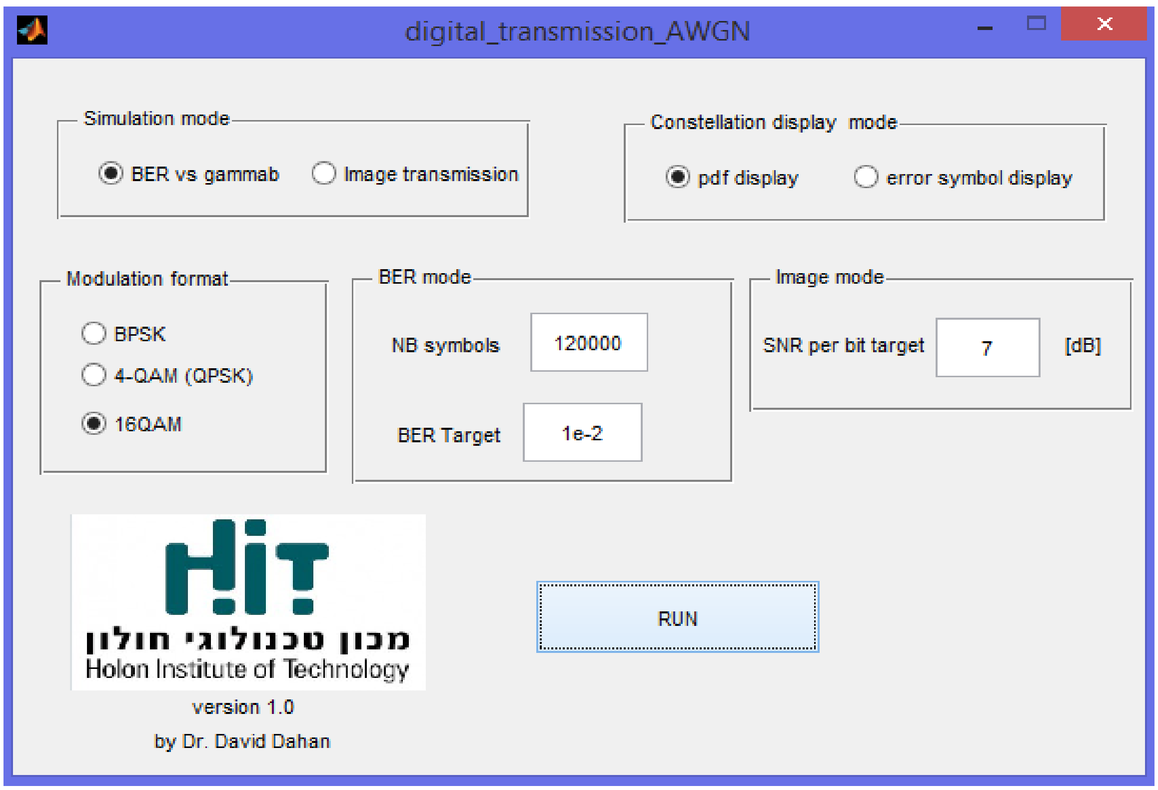
Simulator Objectives

Analyzing the performances of different types of digital modulation formats (BPSK,QPSK and 16QAM) with a discrete AWGN channel model.

Running instructions

The simulator runs in MATLAB environment. Run the function digital_transmission_AWGN.m

Simulator GUI



Simulator parameter configurations

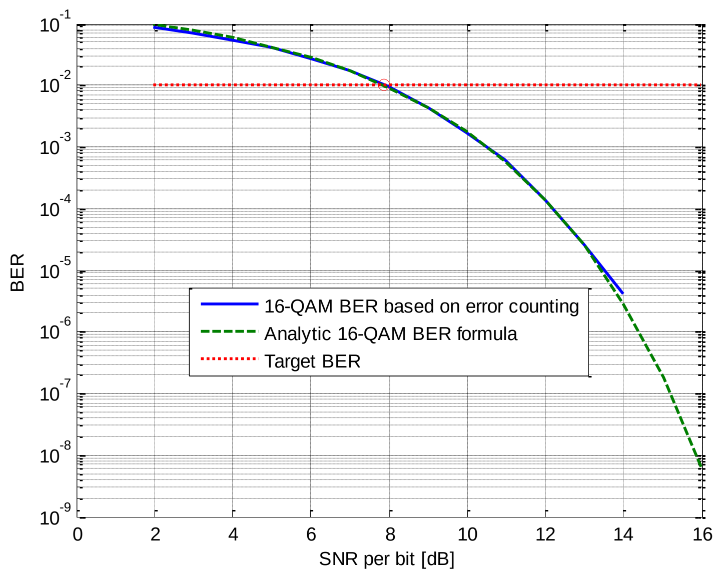
1) Simulation mode

Two simulation mode can be selected :

a) BER mode

In this mode symbols are generated randomly and sent over the discrete AWGN channel . The number of symbols and the BER target are set in the GUI BER mode section.

The simulator displays the constellation diagram for different SNR per bit level (from 16 dB to 0 dB) and resulting BER. At the end of the simulation, a second figure appears which the BER versus SNR per bit curve resulting from the simulation and analytic curve.



In the Matlab prompt the SNR per bit required for the BER target is displayed .

For example :

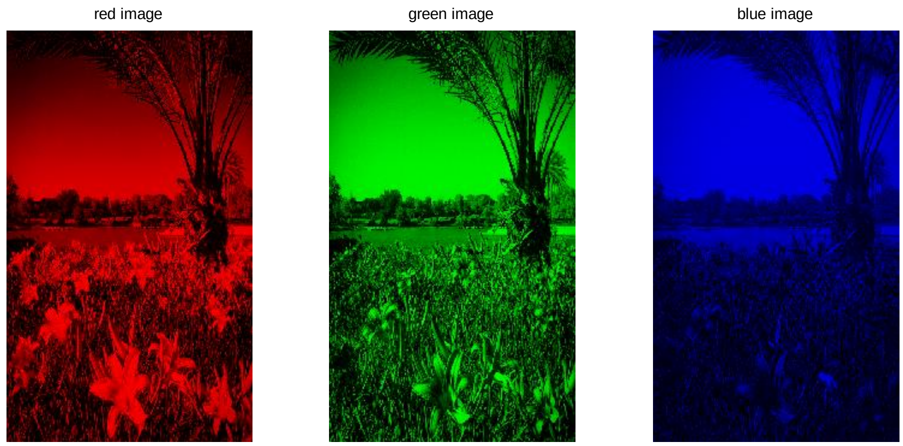
For 16-QAM, the SNR per bit level of 7.8711 dB is required to reach BER=0.01

b) <u>Image mode</u>

In this mode, a colored jpeg image is converted into a bit stream then symbol stream and sent over the discrete AWGN channel. At the receiver the received bit stream is converted back into the image

The image to be sent is the following picture :

An RGB representation : 3 matrixes of unsigned 8-bit integer elements (0-127), each matrix coding one of the 3 color scales (red, green and blue), as shown below

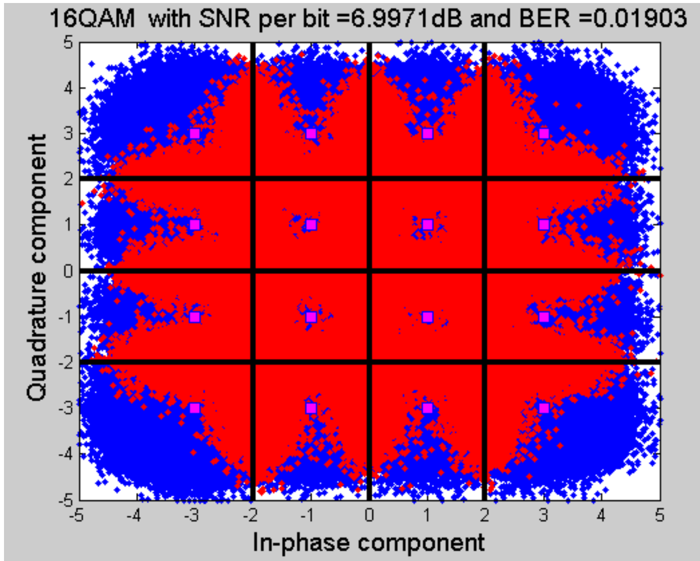


Then each matrix is converted into a bit stream.

The image size is 540 x540 pixels and the bit stream length is 540x540x3x8=6 998 400 bits .

In this mode, the SNR per bit target is set. The transmitted picture and received picture at the SNR per bit target) are displayed

The constellation diagram is also displayed (below example with 16-QAM)

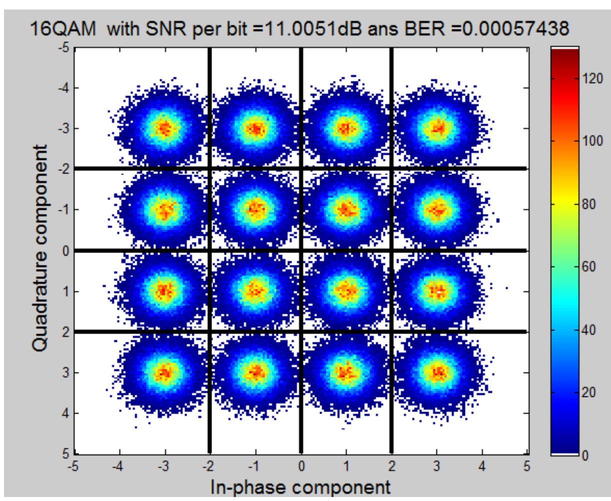


2) Modulation format

3 modulation formats can be selected : BPSK, 4-QAM (QPSK) and 16-QAM

3) Constellation display format
a) pdf display mode

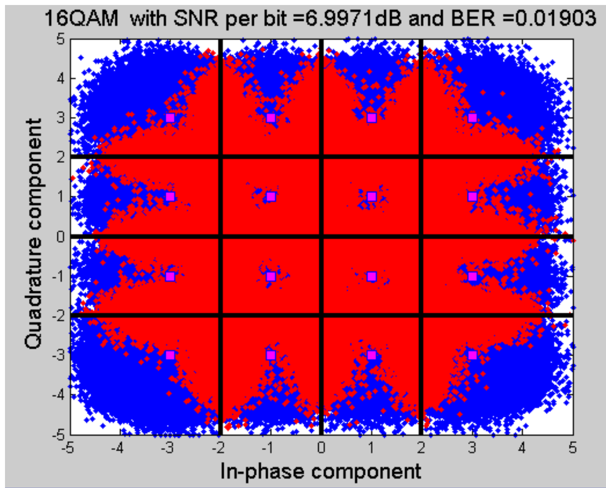
When pdf display mode is selected the constellation diagram is displayed with the pdf isocontour level for around each symbol pdf



Color code : The color bar indicates the color code for the number of symbols located in the given plane coordinates : warn color (high pdf level- high number of symbols) cold color (low pdf level-low number of symbol).

b) error symbol display mode

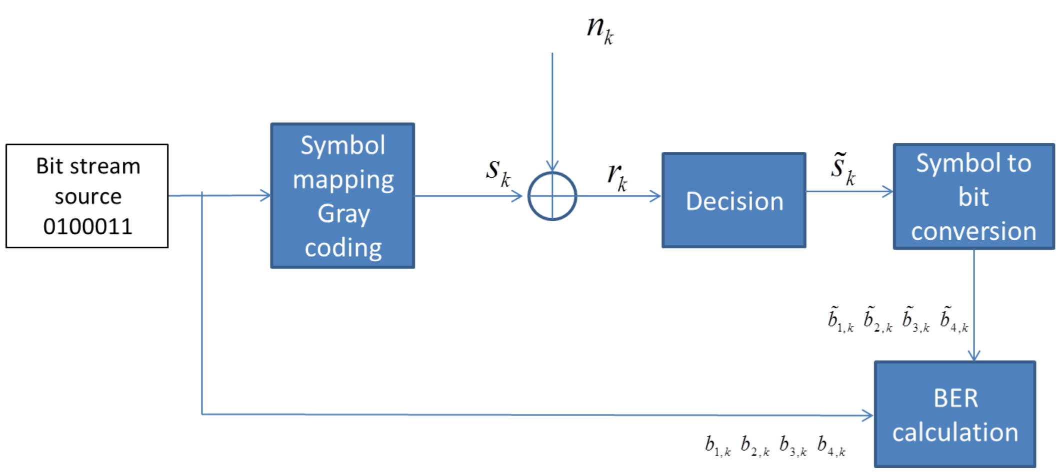
In this mode the transmitted symbol are displayed in purple , the receive noisy symbols that are correctly detected are colored in blue while the noisy received symbols that are not correctly decided are colored in red



3) <u>Channel characteristics</u>

The symbols are is sent over an a discrete model of AWGN (Additive White Gaussian Noise) channel. The discrete AWGN channel is a discrete baseband equivalent channel (discrete complex envelop representation) of the real transmission channel (which is a continuous passband channel)

The k^th received symbol is given by $r_k = s_k + n_k$ where s_k is the k^th sent symbol and the n_k is the k^th noise sampled at the received. n_k is a zero mean Gaussian random variable with variance $N_0/2$



The BER calculation module compares the received bit stream to the sent bit stream.
The BER is calculated by divided the number of error bits by the total number of sent bits